Project Overview Document

1. Introduction

The project is a full-stack e-Commerce web application developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js). The system provides users with an online shopping experience where they can browse products, manage their shopping cart, place orders, and submit product reviews.

The application follows a client-server architecture, where the frontend communicates with a RESTful backend API. Authentication and authorization are implemented using JSON Web Tokens (JWT) to ensure secure access to protected resources.

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2. Project Objectives

The main objectives of the project are:

- To develop a modern and responsive online shopping platform.
- To allow users to browse and purchase products efficiently.
- To provide secure user authentication and authorization.
- To implement cart management functionalities for shopping convenience.
- To enable customers to rate and review products.
- To provide administrators with management capabilities for products and reviews.

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3. Project Scope

The e-Commerce application includes the following core functionalities:

User Features

- User registration and login.
- Product browsing and searching.
- Shopping cart management.
- Product review and rating system.
- Checkout and order placement.

Admin Features

- Product management (Create, Update, Delete).
- User management.

- Review moderation and deletion.
- Order management.

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4. Technologies Used

Frontend

- React.js
- React Router
- Bootstrap
- Formik

Backend

- Node.js
- Express.js
- MongoDB
- Mongoose

Security & Utilities

- JWT Authentication
- Bcrypt Password Hashing
- dotenv
- Morgan
- CORS

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5. Entities and Their Roles

User

Represents customers and administrators using the system. Users can register, log in, manage carts, place orders, and write reviews.

Product

Represents items available for purchase. Products contain details such as name, price, image, and category.

Cart

Stores products selected by a user before checkout. Each cart is associated with one user and contains multiple product items with quantities.

Review

Represents user feedback for products. Reviews include ratings and optional comments linked to both users and products.

Order

Represents completed purchases containing purchased items, shipping details, payment information, and order status.

Category

Organizes products into groups for easier browsing and filtering.